Ozel Yilmazel

he/him/his

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Education

University of Massachusetts Amherst — College of Information and Computer Sciences Amherst, MA

Doctor of Philosophy in Computer Science (First-Year PhD Student) *Present*

Master of Science in Computer Science — GPA: 4.0 *Graduated May 2026*

Bachelor of Science in Computer Science — GPA: 4.0 *Graduated May 2025*

- **Awards and Honors:** Bay-State Fellowship, Phi Kappa Phi Honors, Summa Cum Laude

Work Experience

Center for Intelligent Information Retrieval – UMass Amherst April 2025 – Present

Student Researcher *Amherst, MA*

- Conducted research on cross-domain and multi-domain personalization for retrieval-augmented generation (RAG) systems, as well as long-form generation personalization, publishing a paper that outperformed state-of-the-art methods (SIGIR '26).
- Led MassAITC contract work to develop a novel generative-AI facilitation bot for a new chat application supporting caregivers; the work enabled the grant's success and pioneered the use of generative AI in facilitation agents.
- Currently optimizing retrieval models for AI-agent consumption, addressing how agents ingest data differently and are sensitive to noise while existing search interfaces and interactions remain un-optimized for agentic use.

Initiative for Digital Public Infrastructure – UMass Amherst February 2025 – September 2025

Student Researcher *Amherst, MA*

- Investigated topic distributions on YouTube by applying clustering algorithms to large-scale video embedding datasets (˜16,000 samples). Analyzed how embedding representations influence topic granularity and revealed structural patterns within the platform's content ecosystem.
- Developed a multilingual recommendation analysis pipeline using the YouTube Innertube API, generating recommendation trees to study cross-language exposure paths and user discovery patterns across linguistic boundaries.

Center for Data Science and Artificial Intelligence – UMass Amherst May 2024 – May 2025

Researcher *Amherst, MA*

- Built a machine learning pipeline for song recommendation using learning-to-rank models (XGBoost, scikit-learn) for the music therapy app SingFit, contributing to the associated academic publication.
- Designed and implemented a versatile text classification suite for large language models using LangChain, enabling rapid experimentation across multiple NLP tasks.
- Developed and deployed full-stack web annotation platforms for the NSF Buzzards Bay water preservation project and iNaturalist, leveraging React, PostgreSQL, and Azure cloud services to support environmental and biodiversity research.
- Fine-tuned geo-predictive models with PyTorch, utilizing user-generated labels from iNaturalist, improving species location inference accuracy.

College of Information and Computer Sciences – UMass Amherst September 2022 – Present

Course Teaching Assistant *Amherst, MA*

- Teaching Assistant for CS446 Search Engines: Introduced and taught a new module on Retrieval-Augmented Generation, designed and implemented programming assignments, developed automated grading tools, and supported students through office hours.
- Served as an Undergraduate Course Assistant for CS121, CS230, and CS326, providing academic support to hundreds of students through office hours, project guidance, and grading assistance.
- Received the Outstanding UCA Award for excellence in teaching, communication, and collaboration across multiple semesters.

Projects

OpenResearcher | *Funding and collaboration platform*

- Building a platform that enables academia and industry to find research collaborators, equipped with agentic search functionalities that help industry companies discover labs and academics to fund.
- Developed the full stack with FastAPI and Next.js, including a custom retrieval server offering lexical and semantic retrievers that operate independently or in a hybrid configuration.
- Integrated retrieval with agentic search and automated report generation using LangChain.

Art-GAN | *Generating natural patterns with generative adversarial networks*

- Photographed real-world natural patterns, plants, flowers, and tree bark, and trained a GAN to synthesize new organic patterns, exploring how models perceive the visual structures found in nature.
- Probed the model's intermediate layers and inspected layer activations to surface novel and visually compelling artifacts beyond the final outputs.

Natural Language Playlist Querying | *Generating Spotify playlists with short queries*

- Designed and implemented a natural language playlist retrieval system with a dual-encoder retriever, processing 100,000 Spotify playlists using PyTorch and HuggingFace Transformers.
- Trained custom song embeddings via a word2vec-style objective over playlist track sequences, learning track representations from co-occurrence within playlists.

Publications

Yilmazel, O., Zamani, H., & Allan, J. (2026). Exploring and Improving Cross- and Multi-Domain Personalized Question Answering via a Dual Retrieval Augmentation Approach. In *Proceedings of the 49th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '26)*, 4316–4321. https://doi.org/10.1145/3805712.3809932

Yilmazel, O., et al. Does Retrieval Augmented Generation Collapse When It Retrieves AI-Generated Documents? *GroundLM* (2026), preprint. openreview.net/forum?id=JHwIkYmtxb

Myers, J. R., et al. Developing an equitable machine learning-based music intervention for older adults at-risk for Alzheimer's: Pilot findings for algorithm development and validation. *JMIR Aging*; October 2025; (submitted). (Contributor)

Yilmazel, O., McGrady, R., Partridge, V., & Zuckerman, E. YouTube Topic Modeling. *Computational Social Science Poster Session 2025*; May 2025; UMass Amherst, Amherst. Winner of the People's Choice Award.

Yilmazel, O., Zhu, A., Navarrete, P., Pogorelov, S., & Partridge, V. iNatator: Obtaining Expert Feedback on Species Ranges. *New England Computer Vision Workshop (NECV)*; November 2024; Yale University, New Haven.

Skills

Programming Languages: Python, JavaScript, Java, SQL, HTML/CSS, C#, C

Tools & Frameworks: PyTorch, vLLM, transformers, pandas, NumPy, matplotlib, scikit-learn, NLTK, XGBoost, HuggingFace, LangChain, FastAPI, MongoDB, Unity, PostgreSQL, SLURM, Git, Docker